AI Engineering Roles

Eleven job titles describe the work of building with AI, and most of them overlap. This note ranks them by how often companies hire for them today, says what each one does, and draws the line between the titles that get confused.

What an AI Engineer does day to day, and how the role differs from an ML Engineer, is in the AI Engineer document. This one compares that role with the ten around it.

1. How the roles are ranked

Rank	Role	SimplyHired	Indeed
1	AI Engineer	4,094	2,166
2	Forward Deployed Engineer	3,049	1,868
3	Agentic AI Engineer	378	232
4	AI Solutions Architect	426	221
5	Applied AI Engineer	280	210
6	AI Platform Engineer	149	109
7	MLOps Engineer	122	50
8	Generative AI Engineer	65	28
9	AI Agent Engineer	41	23
10	LLM Engineer	36	12
11	LLMOps Engineer	3	2

Indeed counts by job title only, and it gives exactly this order. Its search can be limited to the title of the posting, so a description that only mentions the role is not counted.

Agentic AI Engineer is above AI Solutions Architect because of growth. SimplyHired puts the architect slightly ahead and Indeed puts the agentic title ahead, so the counts alone do not decide. Agentic AI skills grew by more than 280% in one year, and no comparable figure exists for the architect title.

Published growth supports the top of the list:

Role	Published evidence
AI Engineer	First on LinkedIn's Jobs on the Rise 2026, the fastest-growing title in the United States
Forward Deployed Engineer	Postings up more than 1,000%, January to August 2026 against the same months of 2025 (Lightcast, reported by Fortune)
Forward Deployed Engineer	643 postings on Indeed in April 2025, 5,330 in April 2026 (reported by Business Insider)
Agentic AI Engineer	Agentic AI skills went from 0.06% to 0.23% of US postings between 2024 and 2025, nearly 90,000 postings (Lightcast, Stanford AI Index 2026)

These numbers rank titles, not work. Much MLOps work is advertised as "ML Engineer" or "Platform Engineer", so a low rank means few companies use that name. It does not mean few people do that work.

The "AI Engineer" count includes longer titles. A search for the phrase also finds "Applied AI Engineer", "Generative AI Engineer" and "Agentic AI Engineer". Removing all three, 723 postings, still leaves about 3,370, so first place does not change.

A one-day count is a snapshot. Both boards include reposts from recruitment agencies, and both count postings, not hires. Trust the order more than the exact numbers.

2. The three families

Most of the eleven titles describe one of three jobs. The family tells us what the person is responsible for. Inside one family, the daily work is often the same under different names.

Family	The question it answers	Titles
Build	What does the product do with the model?	AI Engineer, Applied AI, Generative AI, LLM, Agentic AI, AI Agent
Run	Does the model stay fast, cheap and available?	AI Platform, MLOps, LLMOps
Customer	Does it work in this customer's systems?	Forward Deployed Engineer, AI Solutions Architect

A fourth family, the people who train models, is outside this list. ML Engineer and Research Scientist belong there.

Between families, the users differ. Build roles make something for the product's users. Run roles make something for other engineers. Customer roles make something work for one organization.

3. AI Engineer

An AI Engineer is a software engineer who builds product features on top of models that somebody else trained.

In this note the role is the reference point: every other title in the Build family is this role with a narrower scope.

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Skills:  Python, TypeScript or Java · model APIs · prompts · RAG
         evaluation · cost control · ordinary backend engineering
```

It is first because it is the default name. When a company wants someone to add AI features to its product and has no reason to be more specific, this is the title it uses.

4. Forward Deployed Engineer

A Forward Deployed Engineer (FDE) is a software engineer who works inside a customer's organization and builds what is needed to make the product work in that customer's systems.

The title comes from Palantir, which sent engineers to sit with each customer. AI companies adopted it because a model that works in a demonstration still has to be connected to a customer's data, permissions and processes.

Responsibility	What it involves
Find the real problem	Decide with the customer's team what to build first
Integrate	Connect the product to the customer's data, identity system and tools
Build the missing parts	Write the code the standard product does not provide
Deploy and support	Take it to production in the customer's environment
Report back	Tell the product team what customers keep asking for

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Skills:  Python or TypeScript · APIs and data integration · SQL
         cloud and security basics · LLM application patterns
         working directly with customers
```

It writes production code. That separates it from an AI Solutions Architect, who designs and advises but rarely owns the code that ships.

It builds for one customer at a time. An AI Engineer builds for every user of the product. An FDE builds what one customer needs, then passes the repeated needs back to the product team so the next customer gets them as standard.

Unclear problems and time on site are part of the job. Many postings expect travel, and the problem is often not defined when the work starts. Palantir, OpenAI, Anthropic, Google, Microsoft and Meta all advertise the role.

5. Agentic AI Engineer

An Agentic AI Engineer builds systems in which a model plans the steps of a task and calls tools to carry them out, instead of answering once.

The agent loop, tools, MCP and agent memory are explained in the LLM Applications document. This section is about the job.

Responsibility	What it involves
Design the workflow	Decide which steps the model chooses and which stay fixed in code
Build the tools	Expose internal systems as functions or MCP servers, each with narrow permissions
Control the loop	Step limits, token budgets, and approval before any action with side effects
Evaluate whole runs	Score the result of many steps, not one reply
Trace	Record every step, so a wrong action can be explained afterwards

```
Skills:  Python · an agent framework such as LangGraph or Google ADK
         tool calling and MCP · multi-step evaluation · tracing
         permission design
```

The difference from an AI Engineer is side effects. An AI Engineer's feature that is wrong shows a bad answer. An agent that is wrong has already changed something, such as a refund, a ticket or a deployment. Most of this role is the design that limits what the agent can do and checks it before it does it.

6. AI Solutions Architect

An AI Solutions Architect designs how an AI system fits into an organization's existing architecture, and helps choose the models, services and patterns before a team builds it.

Many work for cloud and data platform vendors and design for that vendor's customers. Others work inside a large company and design for its internal teams.

Responsibility	What it involves
Gather requirements	Data sources, users, latency, regulation, budget
Design the system	Model, retrieval, data flow and security boundaries
Choose services	Hosted or self-hosted model, which managed services to use
Estimate cost and risk	Cost per request at expected volume, and what fails first
Prove the design	A proof of concept, then a handover to the team that builds it

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Skills:  cloud architecture on one major provider · LLM and RAG patterns
         security and data governance · cost estimation
         writing and presenting designs
```

It designs, and other people build. An architect produces the design and the proof of concept, then moves to the next customer or project.

Breadth matters more than depth. The architect has to know enough about models, data, networking and security to see where a design will fail. Deep knowledge of one framework matters less.

At a vendor, the role often supports sales. The design shows a customer that the product can do what they need. That is why many postings ask for presentation skills and pre-sales experience.

7. Applied AI Engineer

An Applied AI Engineer uses AI models to solve a specific business or product problem, and is measured by the result in that problem rather than by the model.

Responsibility	What it involves
Define success	Agree on the business number that shows the problem is solved
Choose the approach	A hosted model with a prompt, retrieval, fine-tuning, or a classic ML model
Build and ship	Put the solution into the product
Measure the result	Compare the business number before and after

```
Skills:  Python · model APIs and open models · fine-tuning basics
         classic ML with scikit-learn · experiment design
         knowledge of the business domain
```

In most postings it is the AI Engineer job. When the two titles differ, the applied role goes further into the data and the choice of model. It may pick a small classifier, a model that sorts input into fixed categories, where a large language model would be slower and more expensive.

At some AI labs it is the name of a team. At OpenAI and Anthropic, Applied AI is the group that turns the lab's models into products and customer solutions. Those roles are closer to an FDE or an AI Engineer than to research.

8. AI Platform Engineer

An AI Platform Engineer builds the shared internal platform that other teams use to call, serve and monitor models.

Responsibility	What it involves
Model access	One internal gateway to hosted and self-hosted models
Serving	GPU capacity, autoscaling and inference servers
Shared services	Vector stores, a prompt registry, evaluation runs
Limits and cost	Quotas, API keys, logging, and spend per team
Reliability	SLOs and on-call for everything above

```
Skills:  Kubernetes · Terraform · one major cloud · Go, Python or Java
         inference servers such as vLLM or Triton · observability
         GPU scheduling basics
```

Its users are engineers, not customers. The platform succeeds if product teams can add an AI feature without solving serving, keys, quotas and logging again each time.

It is the general title of the Run family. MLOps and LLMOps describe parts of what an AI platform team does. Larger companies use the platform title and include both in it. The AI gateway itself is explained in the LLM Applications document.

9. MLOps Engineer

An MLOps Engineer automates the lifecycle of machine learning models: training, versioning, deployment and monitoring in production.

Responsibility	What it involves
Training pipelines	Repeatable training runs, from versioned data
Model registry	Every model version stored with its data and its scores
Deployment	CI/CD for models, with a gradual rollout
Monitoring	Prediction quality, latency and drift
Retraining	A new training run when quality drops

```
Skills:  Python · Docker and Kubernetes · CI/CD · MLflow or Kubeflow
         Vertex AI or SageMaker · data pipelines · monitoring
```

It assumes the team trains its own models. Fraud scoring, recommendations and demand forecasts are trained in house on the company's own data. That is the work MLOps was built for.

Drift is the problem the role exists for. Drift means the live data moves away from the data the model was trained on, so the model gets worse without any code change. MLOps detects it and starts a new training run.

10. Generative AI Engineer

A Generative AI Engineer builds features that produce content, such as text, images, audio or code, with generative models.

Responsibility	What it involves
Build generation features	Drafting, summarizing, image or speech generation
Write prompts and templates	Reliable output from a model that varies
Work with several media	Image, speech and video models beside text models
Check the output	Quality evaluation, and filters for unsafe content

```
Skills:  Python · text and image model APIs · prompting · RAG
         evaluation · content safety filters
```

Most postings describe the AI Engineer job. When the title means something different, the difference is media other than text: image, audio and video generation.

11. AI Agent Engineer

An AI Agent Engineer builds and configures individual AI agents for defined tasks, often on a vendor's agent platform.

Responsibility	What it involves
Build the agent	One agent for one task, such as answering order questions
Connect tools	Give it access to the systems it needs, and nothing more
Write its instructions	Behavior, limits and when to hand over to a person
Test and adjust	Run real cases through it and correct what fails

```
Skills:  Python or a low-code agent builder · tool calling
         prompt design · testing with real conversations
```

In most postings it is the Agentic AI Engineer job. Both titles ask for the same skills, and employers use them for each other.

When the titles differ, scope is the difference. An Agentic AI Engineer designs a system of steps, sometimes with several agents, and the controls around it. An AI Agent Engineer builds one agent for one task, and that is often configuration on a CRM or cloud vendor's agent builder rather than code.

12. LLM Engineer

An LLM Engineer works on large language models themselves, as well as on applications that use them: fine-tuning, serving and optimizing the model.

Responsibility	What it involves
Fine-tune	Adapt an open model to a domain, often with LoRA
Serve and optimize	Quantization, batching and inference servers
Evaluate models	Benchmarks and task-specific evaluations
Build on the model	RAG and prompts, as an AI Engineer would

Quantization means storing the model's weights with fewer bits, so it needs less memory and answers faster, with a small loss of quality.

```
Skills:  Python · PyTorch · Hugging Face Transformers · LoRA
         inference optimization · GPU basics · model evaluation
```

It sits between an AI Engineer and an ML Engineer. An AI Engineer takes the model as fixed. An LLM Engineer often changes it by fine-tuning, or runs it on the company's own GPUs. Fine-tuning is explained in the LLM Applications document.

Postings disagree about what the title means. Some describe ordinary application work, and some describe model work close to research. Read the responsibilities: PyTorch, LoRA and GPU work mean the second kind.

13. LLMOps Engineer

An LLMOps Engineer runs language model applications in production: deployment, evaluation, monitoring, cost and safety.

Responsibility	What it involves
Version	Prompts, models and their settings, released like code
Evaluate on each change	A fixed set of test questions run before every release
Monitor	Latency, tokens, cost per request, and answer quality
Control cost	Caching, and routing simple requests to cheaper models
Guard	Prompt injection checks, and filters for personal data

```
Skills:  Python · LLM tracing tools such as Langfuse or LangSmith
         evaluation pipelines · CI/CD · an AI gateway · one major cloud
```

It is MLOps for a model nobody on the team trained. There is no training pipeline and no retraining. The model changes when the provider releases a new version, so the work moves to prompts, evaluation and cost per request.

It is almost never a title on its own. The work appears inside AI Platform, MLOps and AI Engineer postings. Learn the skills, and search for those titles.

14. Telling the roles apart

Three questions place almost any posting, whatever its title:

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Does the person write the production code?
  no   →  AI Solutions Architect
  yes  →  Who uses what they build?
            one customer           →  Forward Deployed Engineer
            other engineers        →  AI Platform, MLOps, LLMOps
            the product's users    →  Does the work change the model?
                                        yes  →  LLM Engineer
                                        no   →  AI Engineer and its variants
```

Read the responsibilities, not the title. Titles in this area change every year, and the same title means different work at two companies. The responsibilities section of a posting answers the three questions above in a few lines.

The pieces together

Role	In one line
AI Engineer	Builds product features on models somebody else trained
Forward Deployed Engineer	Makes the product work inside one customer, and writes the code
Agentic AI Engineer	Builds systems where the model chooses steps and takes actions
AI Solutions Architect	Designs the system and hands it to the team that builds it
Applied AI Engineer	Solves one business problem, with whatever model fits
AI Platform Engineer	Gives other teams one way to call, serve and monitor models
MLOps Engineer	Trains, deploys, watches and retrains in-house models
Generative AI Engineer	An AI Engineer, sometimes with images, audio or video
AI Agent Engineer	Builds one agent for one task, often on a vendor platform
LLM Engineer	Fine-tunes, serves and optimizes the model itself
LLMOps Engineer	Runs model applications in production: evaluation, cost, safety

For example, a software company adds an AI support assistant to its product. An AI Engineer builds the feature that answers customers from the help articles. An Agentic AI Engineer lets it issue refunds through a tool that needs a person's approval above a set amount.

The AI Platform team runs the gateway that every model call goes through, and does the LLMOps work: an evaluation run on every prompt change, and a cost report per team. An LLM Engineer fine-tunes a small open model to sort incoming tickets, because the large model is too expensive for that volume. The MLOps Engineer keeps retraining the existing churn model, which has nothing to do with language.

A bank wants the assistant but requires its data to stay in its own cloud, so an AI Solutions Architect designs that setup and a Forward Deployed Engineer spends six weeks at the bank connecting it to their ticket system. Two months later the model provider releases a new version and the bank's answers get worse. The evaluation run fails before the change reaches the bank, and the release waits.

The titles change faster than the work. Three things stay constant: who uses what we build, whether we change the model, and whether we write the code ourselves.